13.4: Genetic variation produced in sexual life cycles contributes to evolution

How do we account for the genetic variation of the family members in Figure 13.1? As you will learn more about in later chapters, mutations are the original source of genetic diversity. These changes in an organism's DNA create the different versions of genes, known as alleles. Once these differences arise, reshuffling of the alleles during sexual reproduction produces the variation that results in each member of a sexually reproducing population having a unique combination of traits.

Origins of Genetic Variation Among Offspring

In species that reproduce sexually, the behavior of chromosomes during meiosis and fertilization is responsible for most of the variation that arises in each generation. Three mechanisms contribute to the genetic variation arising from sexual reproduction: independent assortment of chromosomes, crossing over, and random fertilization.

Independent Assortment of Chromosomes

One aspect of sexual reproduction that generates genetic variation is the random orientation of pairs of homologous chromosomes at metaphase of meiosis I. At metaphase I, the homologous pairs, each consisting of one maternal and one paternal chromosome, are situated at the metaphase plate. (Note that the terms *maternal* and *paternal* refer, respectively, to whether the chromosome in question was contributed by the mother or the father of the individual whose cells are undergoing meiosis.) Each pair may orient with either its maternal or paternal homolog closer to a given pole—its orientation is as random as the flip of a coin. Thus, there is a 50% chance that a particular daughter cell of meiosis I will get the maternal chromosome of a certain homologous pair and a 50% chance that it will get the paternal chromosome.

Because each pair of homologous chromosomes is positioned independently of the other pairs at metaphase I, the first meiotic division results in each pair sorting its maternal and paternal homologs into daughter cells independently of every other pair. This is called *independent assortment*. Each daughter cell represents one outcome of all possible combinations of maternal and paternal chromosomes. As shown in **Figure 13.11**, the number of combinations possible for daughter cells formed by meiosis of a diploid cell with two pairs of homologous chromosomes (n=2) is four: two possible arrangements for the first pair times two possible arrangements for the second pair. Note that only two of the four combinations of daughter cells shown in the figure would result from meiosis of a *single* diploid cell because a single parent cell would have one or the

other possible chromosomal arrangement at metaphase I, but not both. However, the population of daughter cells resulting from meiosis of a large number of diploid cells contains all four types in approximately equal numbers. In the case of n=3, eight combinations (2×2×2=23) of chromosomes are possible for daughter cells. More generally, the number of possible combinations when chromosomes sort independently during meiosis is 2n, where n is the haploid number of the species.

Figure 13.11

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The independent assortment of homologous chromosomes in meiosis.
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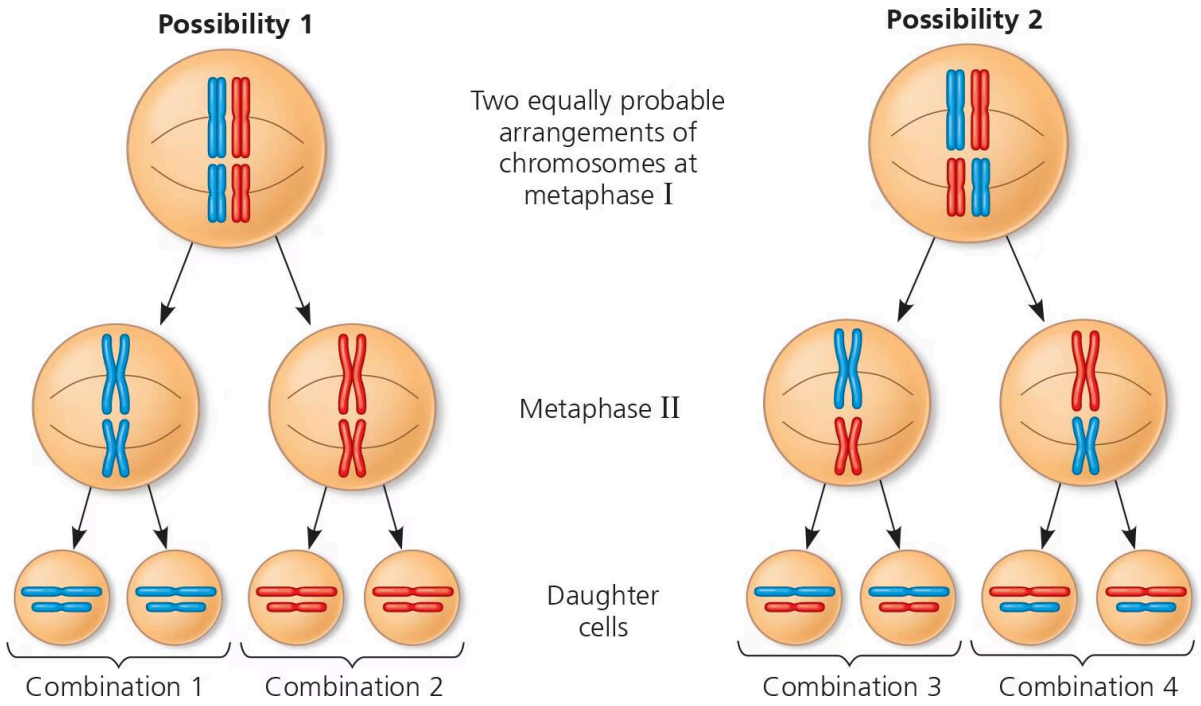


Two equally probable arrangements of chromosomes at metaphase 1 are shown with the resulting daughter cells (Possibility 1 and Possibility 2). There are four chromosomes in two homologous pairs (a long homologous pair and a short homologous pair) present in the parent cell.

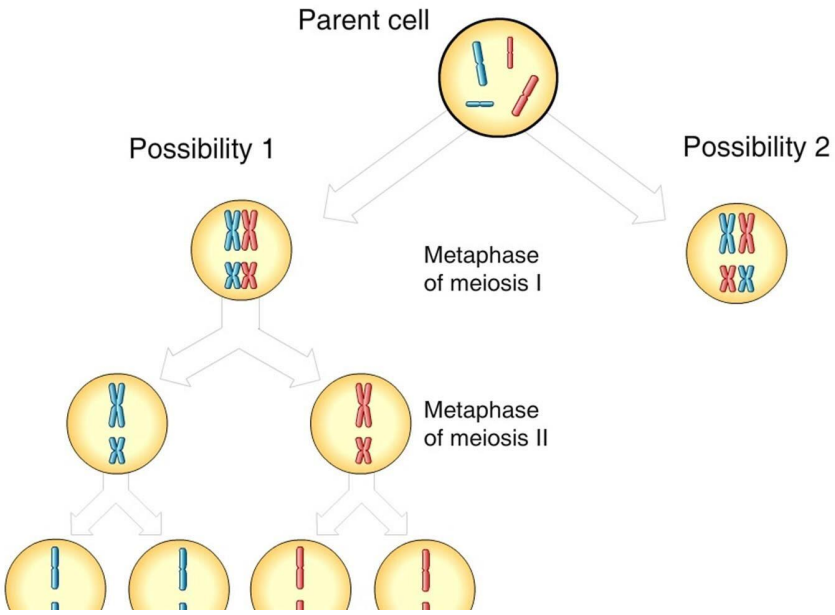
- Possibility 1 shows the homologous pairs oriented at Metaphase 1 as follows: duplicated long paternal chromosome oriented with duplicated short paternal chromosome and duplicated long maternal chromosome oriented with duplicated short maternal chromosome.
- Following Meiosis 1 (homologous pairs separate), two haploid daughter cells are shown in Metaphase 2 (sister chromatids remain attached at the centromeres) with one cell containing two duplicated paternal chromosomes (one long and one short) and one cell with two duplicated maternal chromosomes (one long and one short).
- Following Meiosis 2 (sister chromatids separate) the four haploid daughter cells that result contain long paternal and short paternal chromosomes (2) and long maternal and short maternal

chromosomes (2).

- Possibility 2 shows the homologous pairs in Metaphase 1 oriented as follows: duplicated long paternal chromosome oriented with duplicated short maternal chromosome and duplicated long maternal chromosome oriented with duplicated short paternal chromosome.
- After Meiosis 1 (homologous pairs separate and daughter cells are haploid) one daughter cell contains duplicated long paternal and duplicated short maternal chromosomes, and the other daughter cell contains the duplicated long maternal and short paternal chromosomes. Both daughter cells shown in Metaphase 2.
- After Meiosis 2 (sister chromatids separate) the four haploid daughter cells contain a long paternal and short maternal chromosome (2) and a long maternal and short paternal chromosome (2).

Animation

Genetic Variation from Independent Assortment of Chromosomes



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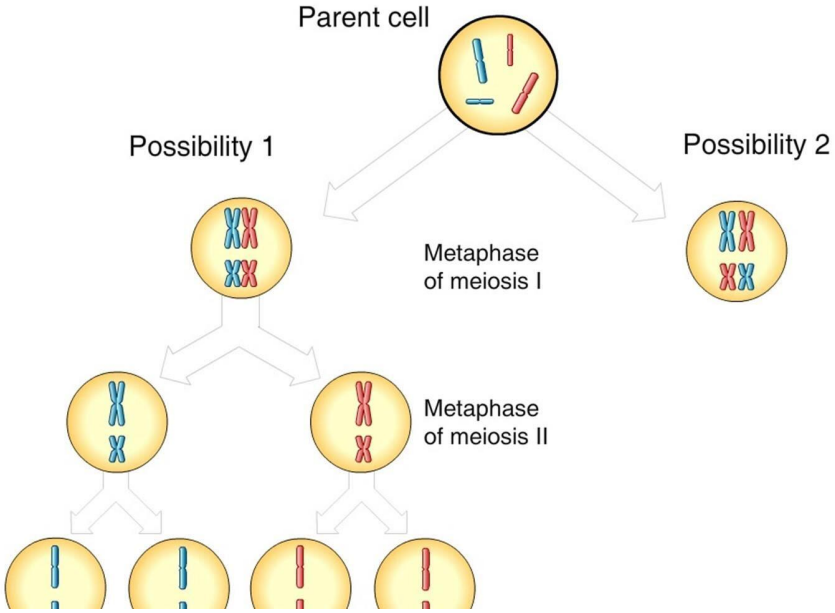

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In the case of humans (n=23), the number of possible combinations of maternal and paternal chromosomes in the resulting gametes is 223, or about 8.4 million. Each gamete that you produce in your lifetime contains one of roughly 8.4 million possible combinations of chromosomes. This is an underestimate, because it doesn't take into account crossing over, which we'll consider next.

Crossing Over

As a consequence of the independent assortment of chromosomes during meiosis, each of us produces a collection of gametes differing greatly in their combinations of the chromosomes we inherited from our two parents. Figure 13.11 suggests that each chromosome in a gamete is exclusively maternal or paternal in origin. In fact, this is *not* the case, because crossing over produces **recombinant chromosomes**, individual chromosomes that carry genes (DNA) from two different parents **(Figure 13.12)**. In meiosis in humans, an average of one to three crossover events occurs per chromosome pair, depending on the size of the chromosomes and the position of their centromeres.

Figure 13.12

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The results of crossing over during meiosis.
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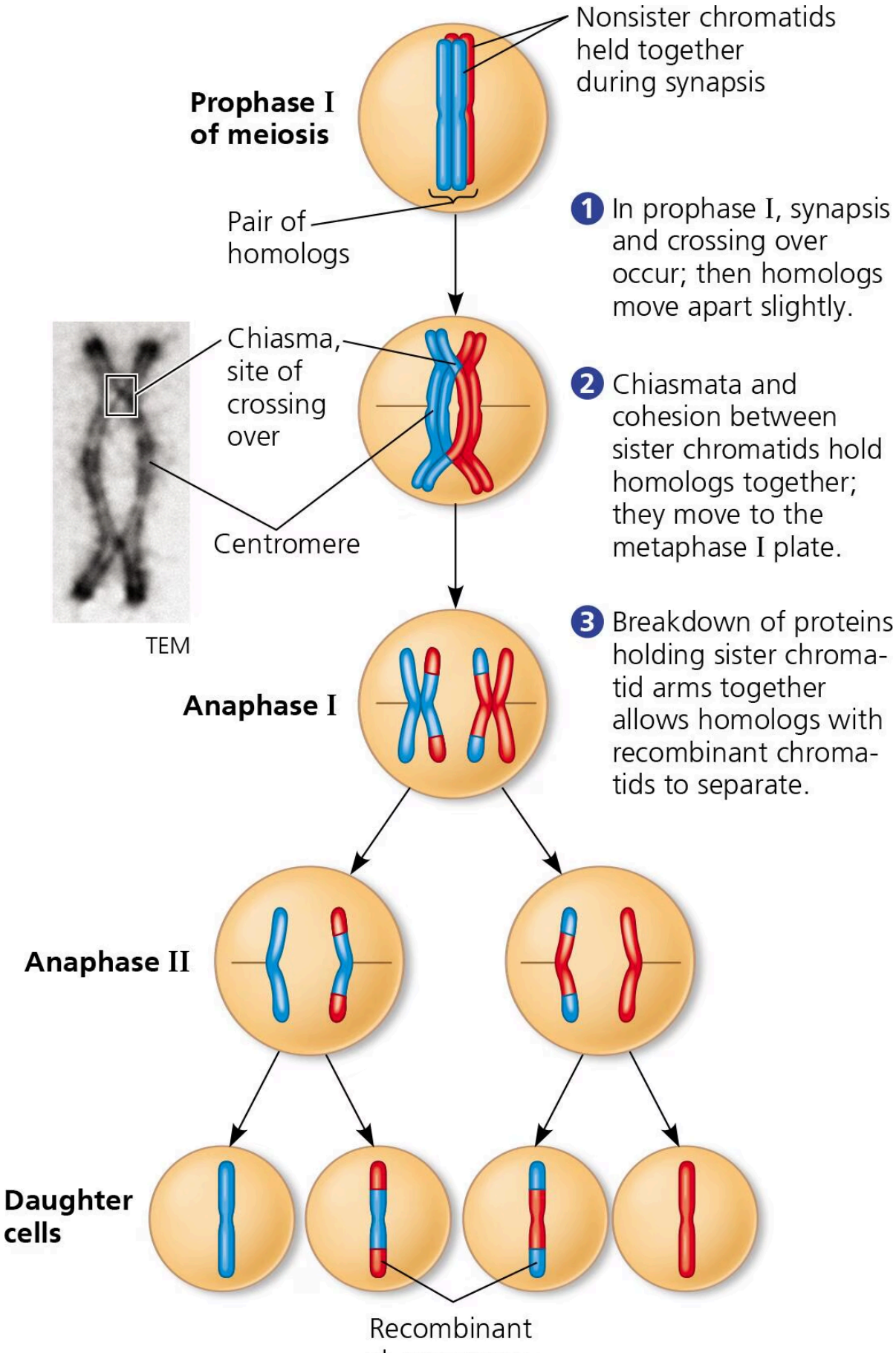

Nonsister chromatids held together during synapsis
Prophase I of meiosis
Pair of homologs
1 In prophase I, synapsis and crossing over occur; then homologs move apart slightly.
Chiasma, site of crossing over
Centromere
TEM
2 Chiasmata and cohesion between sister chromatids hold homologs together; they move to the metaphase I plate.
3 Breakdown of proteins holding sister chromatid arms together allows homologs with recombinant chromatids to separate.
Anaphase I
Anaphase II
Daughter cells
Recombinant

chromosomes

The result of crossing over during meiosis is shown. At Prophase 1 homologous pairs line up and non-sister chromatids are held together during synapsis.

Step 1: In prophase 1, synapsis and crossing over occur; then homologs move apart slightly.

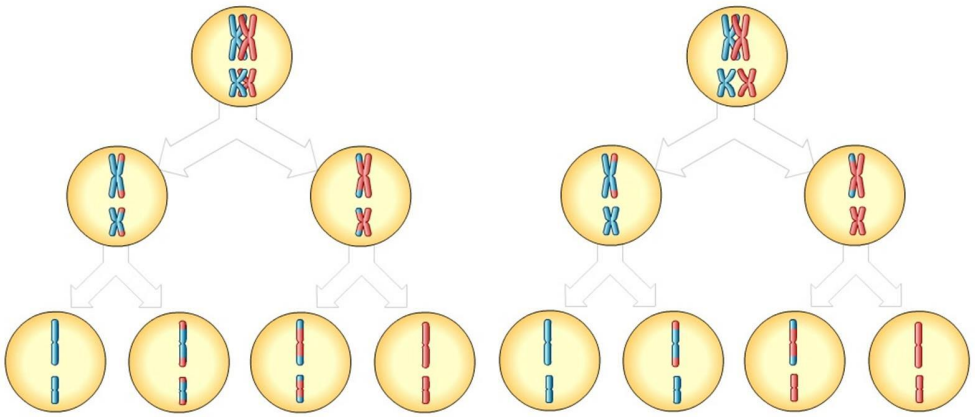
Step 2: Chiasmata and attachments between sister chromatids hold homologs together; they move to the metaphase 1 plate. Chiasma, the site of crossing over, is shown both in a diagram as well as T E M view. The centromere is also shown.

Step 3: Breakdown of proteins holding sister chromatid arms together allows homologs with recombinant chromatids to separate. The results of crossing over are drawn here by showing small bits of maternal chromosome (red) now on one paternal chromosome and small bits of paternal chromosome (blue) now on a maternal chromosome, creating recombinant chromosomes.

Anaphase 2 I is shown with 2 cells with 2 chromosomes each – one cell with a paternal and recombinant chromosome and one cell with a recombinant and maternal chromosome. Following Meiosis 2 the four daughter cells are shown – one with a paternal chromosome, one with a maternal chromosome, and two recombinant chromosomes.

Animation

Genetic Variation from Crossing Over



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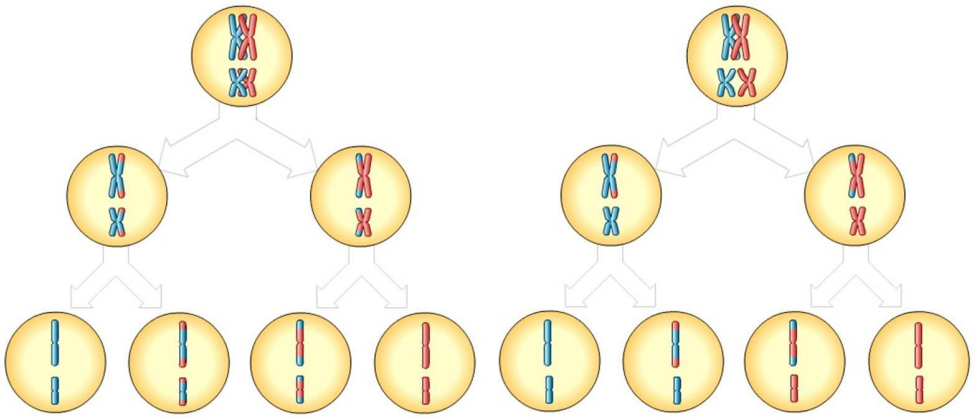

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As you learned in Figure 13.9, crossing over produces chromosomes with new combinations of maternal and paternal alleles. At metaphase II, chromosomes that contain one or more recombinant chromatids can be oriented in two alternative, nonequivalent ways with respect to other chromosomes because their sister chromatids are no longer identical (see Figure 13.12). The different possible arrangements of nonidentical sister chromatids during meiosis II further increase the number of genetic types of daughter cells that can result from meiosis.

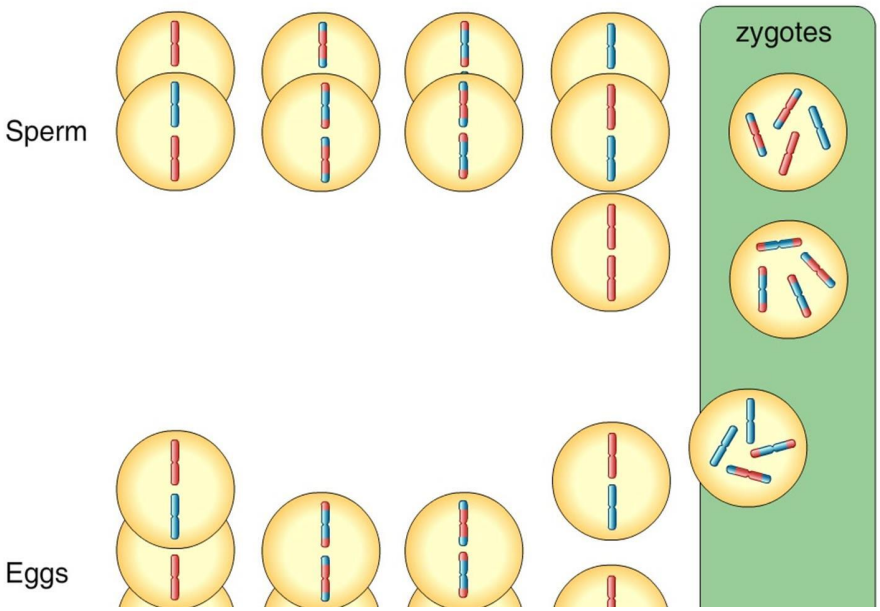
You'll learn more about crossing over in Concept 15.3. The important point for now is that crossing over, by combining DNA inherited from two parents into a single chromosome, is an important source of genetic variation in sexual life cycles.

Random Fertilization

The random nature of fertilization adds to the genetic variation arising from meiosis. In humans, each male and female gamete represents one of about 8.4 million (223) possible chromosome combinations due to independent assortment. The fusion of a male gamete with a female gamete during fertilization will produce a zygote with any of about 70 trillion (223×223) diploid combinations. If we factor in the variation brought about by crossing over, the number of possibilities is truly astronomical. You really *are* unique.

Animation

Genetic Variation from Random Fertilization



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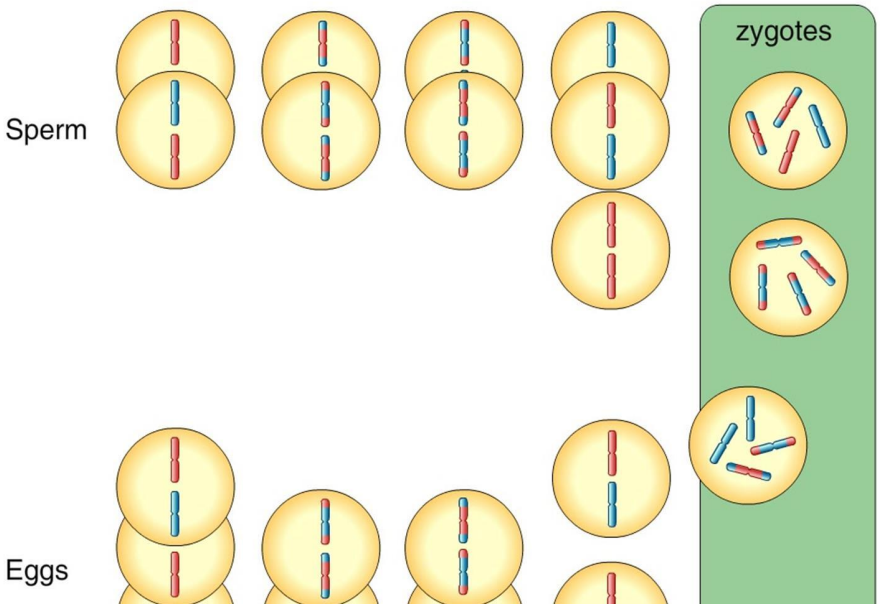

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The Evolutionary Significance of Genetic Variation Within Populations

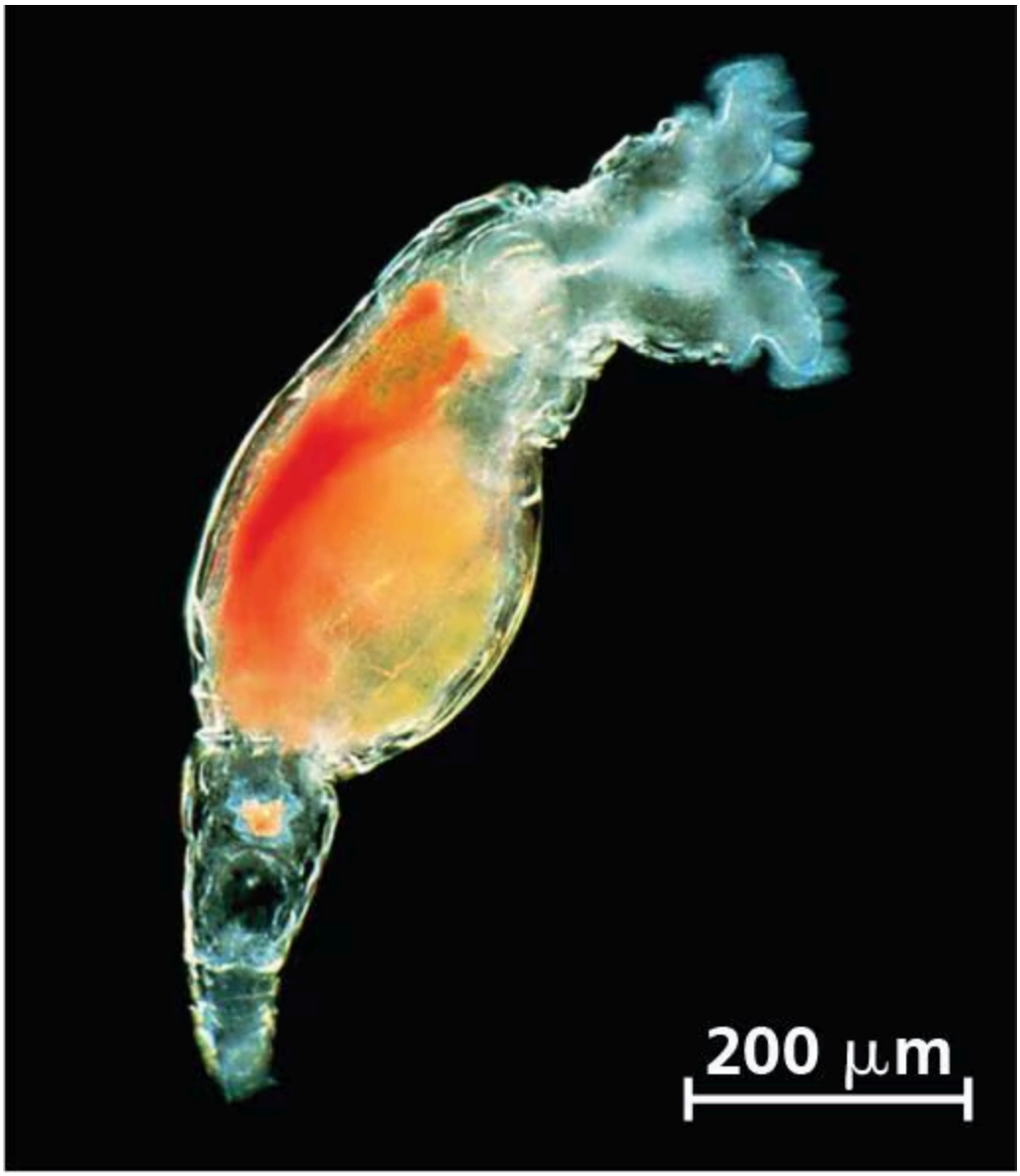
EVOLUTION Now that you've learned how new combinations of genes arise among offspring in a sexually reproducing population, how does the genetic variation in a population relate to evolution? Darwin recognized that a population evolves through the differential reproductive success of its variant members. On average, those individuals best suited to the local environment leave the most offspring, thereby transmitting their genes. Thus, natural selection results in the accumulation of genetic variations favored in a certain environment. As the environment changes, the population may survive if, in each generation, at least some of its members can cope effectively with the new conditions. Mutations are the original source of different alleles, which are then mixed and matched during meiosis. New and different combinations of alleles may work better than those that previously prevailed.

In a stable environment, though, sexual reproduction seems as if it would be less advantageous than asexual reproduction, which ensures perpetuation of successful combinations of alleles. Furthermore, sexual reproduction is more expensive energetically than asexual reproduction. In spite of these apparent disadvantages, sexual reproduction is almost universal among animals. Why is this?

The ability of sexual reproduction to generate genetic diversity is the most commonly proposed explanation for the evolutionary persistence of this process. However, consider the unusual case of the bdelloid rotifer **(Figure 13.13)**. It appears that this group may not have reproduced sexually for more than 50 million years of their evolutionary history, a model that has been supported by recent analysis of the genetic sequences in its genome. Does this mean that genetic diversity is not advantageous in this species? It turns out that bdelloid rotifers are an exception to the "rule" that sex alone generates genetic diversity: Bdelloids have mechanisms other than sexual reproduction for generating genetic diversity. For example, they live in environments that can dry up for long periods of time, during which they can enter a state of suspended animation. In this state, their cell membranes may crack in places, allowing entry of DNA from other rotifer species and even from more distantly related species. Evidence suggests that this foreign DNA can become incorporated into the genome of the bdelloid, leading to increased genetic diversity. In fact, the genomic analysis shows that bdelloid rotifers pick up non-bdelloid DNA at a much higher rate than most other species pick up foreign DNA. The conclusion that bdelloid rotifers have developed other ways of achieving genetic diversity supports the idea that genetic diversity is advantageous, but points out as well that sexual reproduction is not the only way of generating such diversity.

Figure 13.13

A bdelloid rotifer, an animal that reproduces only asexually.



In this chapter, we have seen how sexual reproduction greatly increases the genetic variation present in a population. Although Darwin realized that heritable variation is what makes evolution possible, he could not explain why offspring resemble—but are not identical to—their parents. Ironically, Gregor Mendel, a contemporary of Darwin, published a theory of inheritance that helps explain genetic variation, but his discoveries had no impact on biologists until 1900, more than 15 years after Darwin (1809–1882) and Mendel (1822–1884) had died. In the next chapter, you'll learn how Mendel discovered the basic rules governing the inheritance of specific traits.